MATH 213 Homework 2

Reading

Read Section 1.2 as far as the end of the first paragraph in subsection 1.2.3 of our textbook linked here.

Much of the terminology in this section should be familiar to you from class on Friday. Do pay close attention to subsection 1.2.2 where the authors discuss ordinal and nominal categorical variables, and they also break numerical variables into two further categories: continuous and discrete. This distinction is important for one of the homework exercises.

Written Homework

Type your answers to the homework questions directly into this document in the appropriate spaces. When you are done, SAVE the document as a PDF file and upload it in Blackboard. (Please do not submit other file formats, thank you!!)

1. Work Question 16 "Shows on Netflix" from Chapter 1 of our textbook, linked here.
   BEFORE BEGINNING, identify the CASES for the visualization.
   a) List the variables necessary to create this visualization
      1) release decade
      2) Production country
      3) Rating

   b) **Classify each variable as numerical or categorical**

   1) Release decade (type: Categorical)

   2) Production country (type: categorical)

   3) rating (type: categorical)

   Note that the textbook has solutions to the odd-numbered problems at the end of the book. If you are unsure about how to answer a textbook homework problem, you can look at a similar odd-numbered problem and check its solutions for some ideas about what to do.

2. The following link will take you to the **"code book"** for a data set called "mariokart" compiled by our textbook's authors: https://www.openintro.org/data/index.php?data=mariokart. The code book explains how the data set was gathered and what the cases and variables are.

   You can look at the data set itself in a spreadsheet program using the links at the bottom of that page. I have also loaded the data at the following online spreadsheet: https://docs.google.com/spreadsheets/d/1k4vYL2-6XHA2VhrUhxhvpY5ak9Z0zMz5Gm-3fAB-jFg/edit?usp=sharing

a. Identify the cases for this study. (You will need to look at the code book!)
   the cases are the individual Ebay auctions for the game, each row in the dataset will represent a specific auction

b. List four variables recorded about the cases in this study, using the real-world description from the code book and not just the short heading in the data matrix. Identify the variables as numerical or categorical.

Variable 1: total price

Type (categorical or numerical): numerical

Variable 2: condition of the item

Type: categorical

Variable 3: number of bids

Type: numerical

Variable 4: shipping speed

Type: categorical

c. Pose two research questions – one based on a single variable in the data set and a second that relates to two or more variables in the data set.

1) what is the average total price of Mario kart auctions?

(one var: total price)

2) is there a relationship between the condition of the item and total price of the item at auction?

(two var: condition of item, total price)

3. **[Adapted from our textbook] Sinusitis and antibiotics.** Researchers studying the effect of antibiotic treatment for acute sinusitis compared to symptomatic treatments randomly assigned 166 adults diagnosed with acute sinusitis to one of two groups: treatment or control. Study participants received either a 10-day course of amoxicillin (an antibiotic) or a placebo similar in appearance and taste. The placebo consisted of symptomatic treatments such as acetaminophen, nasal decongestants, etc. At the end of the 10-day period, patients were asked if they experienced improvement in symptoms.(Garbutt et al. 2012)
   a. Who or what are the cases in this study?
      The 166 adults diagnosed with acute sinusitis (each participant)
   b. What variables were recorded for the cases in the study? Classify each variable as categorical or numerical.

1) Treatment group: type -> categorical
2) Improvement in symptoms: type: categorical
3) Age of participant: type -> numerical
4) Duration of symptoms before treatment: type -> numerical